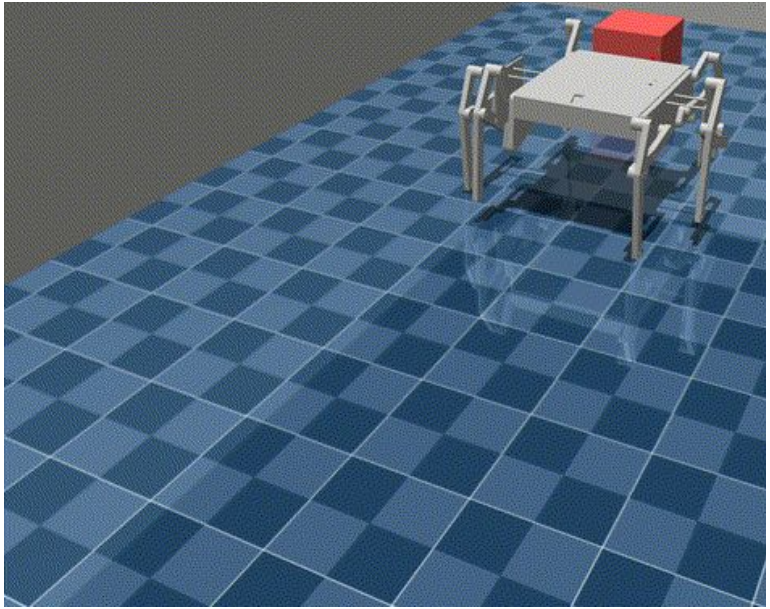


Dance Hexapod Navigation

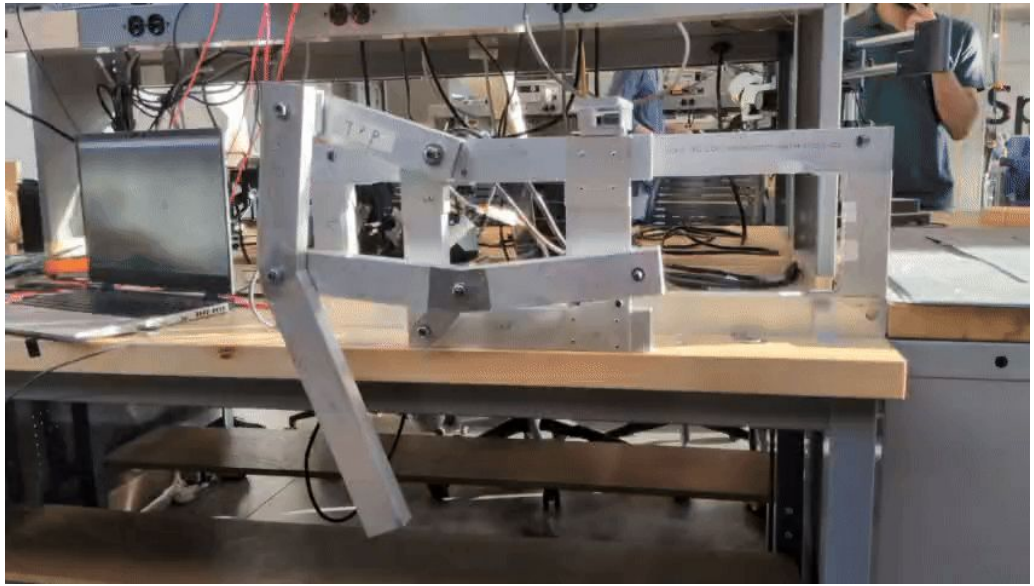
Daniel González Lázaro

4/20/2026

I am controlling a hexapod to achieve curvilinear tripod navigation with strict hardware constraints

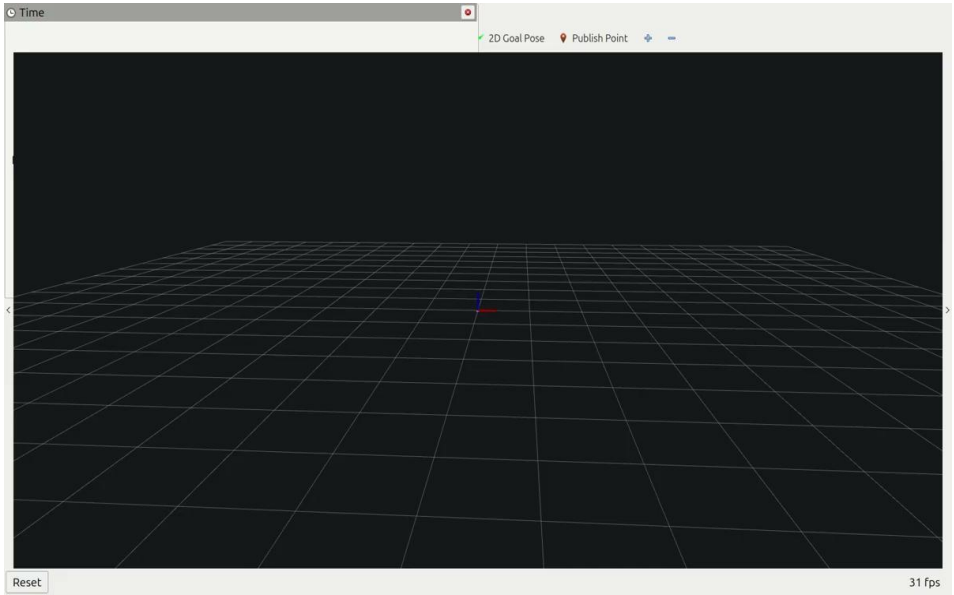
The robot must achieve navigation goals within 6'' error tolerance in varying light, occlusion environments without powered/encoding support wheels

Along with tuned control, linkage kinematics have to be modeled in order to estimate body odometry from drifting encoder measurements



I attempt to reduce uncertainty of mechanical phase by measuring and discretizing ground contact force correlated with specific mechanical phases

I am testing behavior of camera-based slam systems based on tolerance of dynamic environments



The robot must maintain proper ground-truth arc movement. I evaluate the effect of odometry strategy, low-cost inertia measurement and camera input.